Sustainable Energy Systems Portfolio
Fall Interim Report

SESP-INT-IRP

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1.0: Introduction

The purpose of this document is to outline the work I have done as part of the integration team during the fall 2023 semester for the SESP capstone project. The project began with lots of time spent defining the scope, deciding on sub-teams, and the best course to achieve this year's goals. Since TM-1, progress increased as each sub-team's and the overall project's goals became more defined. Most of the work I have conducted has been on projecting a 2050 EV Load Profile, as well as building a "dummy model" that will provide the building blocks of the optimization model that the integration team will work on during the winter semester.

Section 2 of this report describes the methodology and assumptions made to project the 2050 EV Load Profile. Section 3 outlines the initial optimization model. All data used in this report can be found in Appendix A, while Appendix B contains the code referenced in this report.

2.0: 2050 EV Projections

This section introduces the methodologies and assumptions guiding the projection of the 2050 EV Load Profile. Section 2.1 discusses three primary EV charging profiles: residential, workplace, and weekend—with data sourced from Ottawa and San Diego. Section 2.2 focuses on projecting the total number of EVs for 2050, considering factors like population growth, vehicle lifespan, and market trends. The impact of temperature on EV energy consumption is analyzed in Section 2.3, where several studies have been referenced to validate a consumption model across different temperature ranges. In Section 2.4, the discussion shifts to commute distances in Ottawa, using survey data to calculate daily driving patterns. Finally, Section 2.5 describes the use of a MATLAB script to model the hourly EV demand for 2050, incorporating inputs like the number of EVs, charging behaviour, and demand projections.

2.1: Current EV charging profiles

Currently, three EV charging profiles are used: residential, workplace, and weekend. Currently, the residential and weekend profiles come from data collected in Ottawa. The workplace charging profiles originate from data collected in San Diego. As EV adoption increases, one potential change is an increased use of workplace/public charging. Current research suggests that 80-90% of consumers primarily use Residential charging [1]. The three charging profiles being used are shown in Figure 1 below.

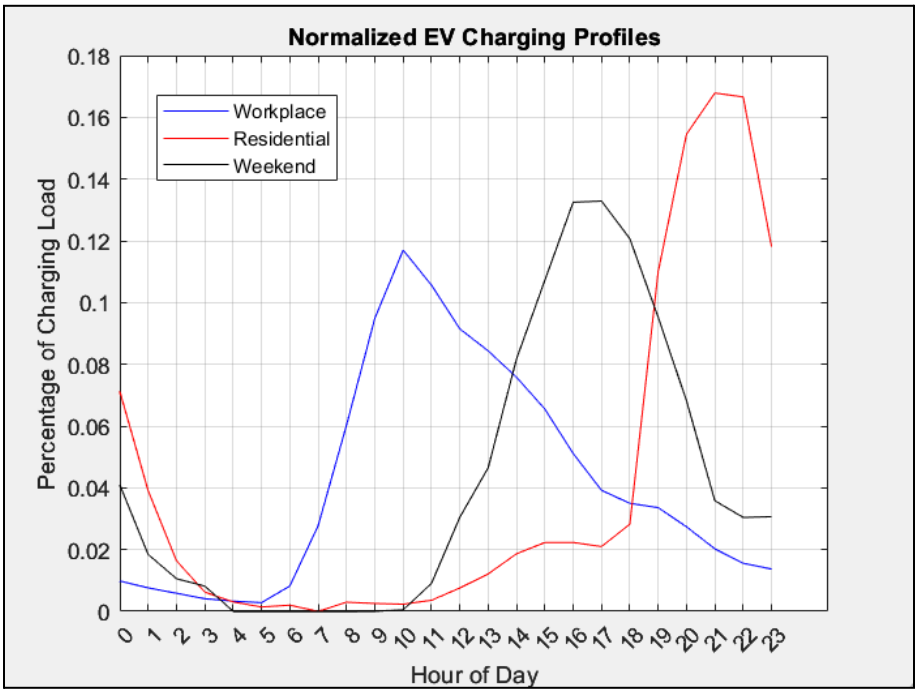


Figure 1: Normalized EV charging profiles.

In the current projections, I assumed that during holidays, EVs will use a weekend profile. Further research on this could be useful to find how driving habits change on holidays such as Christmas.

2.2: Projecting the Total Number of EVs

To project the 2050 total EV load, three main assumptions had to be made. Firstly, population growth had to be estimated. Using projected population growth (2018-2046) [1], I applied the trend to find annual population growth from (2046-2050).

Secondly, I projected the number of new cars on the road each year due to population growth based on the 2013 population (953,148) and the number of registered passenger vehicles in Ottawa in 2013 (463,370) [2]. I calculate the number of vehicles per capita to be 0.486. The number of new vehicles on the road each year is calculated based on the increase in population that year, multiplied by cars per capita (0.486).

I assume the average time a vehicle spends on roads in Canada to be 15.4 years [3]. After a vehicle retires, I assume the driver purchases a new car.

Third, I estimated the number of new EVs each year. For this, I used the projected S-curve for EV market share in Ontario [4]. The City of Ottawa plans for 100% of new car sales to be net zero by 2040. This assumption was held for the calculation done. Figure 2 below shows the projected EV market share for Ontario until 2040, at which point I assume 100% of passenger vehicle sales to be EV.

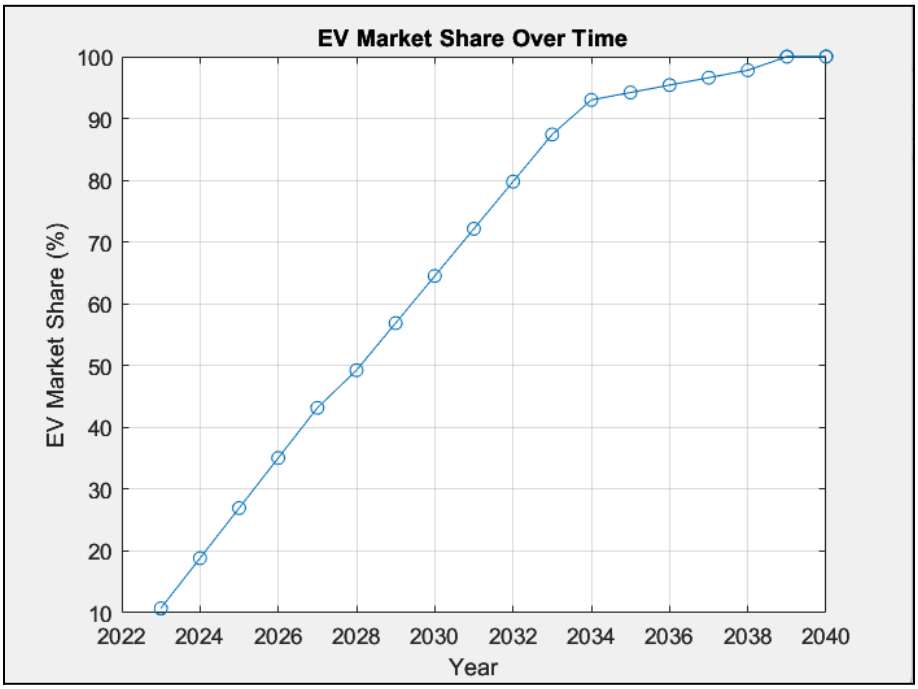


Figure 2: EV Market Share Over Time

The number of new EVs each year is a product of the EV market share and the total number of new cars that year. The number of new cars will be the sum of new cars that year due to population growth and the number of vehicles that retired that year (cars bought 15 years previously).

Finally, the number of EVs on the road each year is the total of EV car sales in the previous 15 years.

I estimate the number of electric vehicles on the road in 2050 to be 412,000. Additionally, the projected number of EVs on the road each year is shown in the appendix of this report.

2.3: Temperature Effects

To find the effect temperature has on the energy consumption of an EV, I followed the same methodology as outlined in [5]. The paper describes a method of finding the energy consumption of the Nissan Leaf as a function of temperature. Using MATLAB, I converted the consumption as a function of miles and temperature (Fahrenheit) to km and Degrees Celsius. I show these two functions in Figure 3 below.

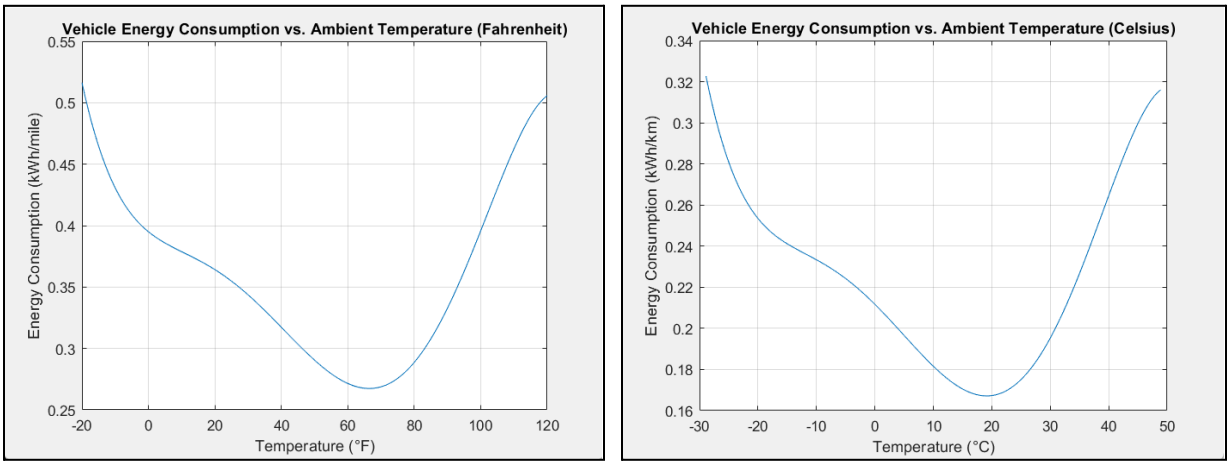


Figure 3: Energy consumption as a function of temperature

The energy consumption used in the current projections follows this profile and is a function of the average temperature on that day. I assumed that all vehicles follow this same relationship. Concerns were raised about the validity of the above energy consumption-temperature relationship because it was collected only using older models of the Nissan Leaf.

After further research to either validate this data or find new data, I have determined that this consumption profile is a good estimate for modern EVs. A study from Munich found that the maximum increase in energy consumption for the Tesla Model 3 and BMW I3 was 50% in cold conditions [6]. This is due to the increased heating load during winter, as well as the decreased battery capacity, along with other factors. Another study was also found to help validate the data used. The study [7], found that at -20°, EVs saw an increase in energy consumption of 44%. The data used found that from the most efficient temperature (18 degrees Celsius) to -20° the Nissan Leaf also experienced an increase in energy consumption of 44%.

I collected data for the past ten years from [8]. I found the hourly temperature and the average of these hourly values for each hour of the year (omitting leap days). I used data from [9] to project the impacts of climate change. It forecasted a 2-degree median temperature change by 2050. I added this two-degree increase to the average hourly temperature. When modelling, all EVs use the energy consumption that corresponds to each hour's temperature. The current temperature profile can be found in Figure 4 below.

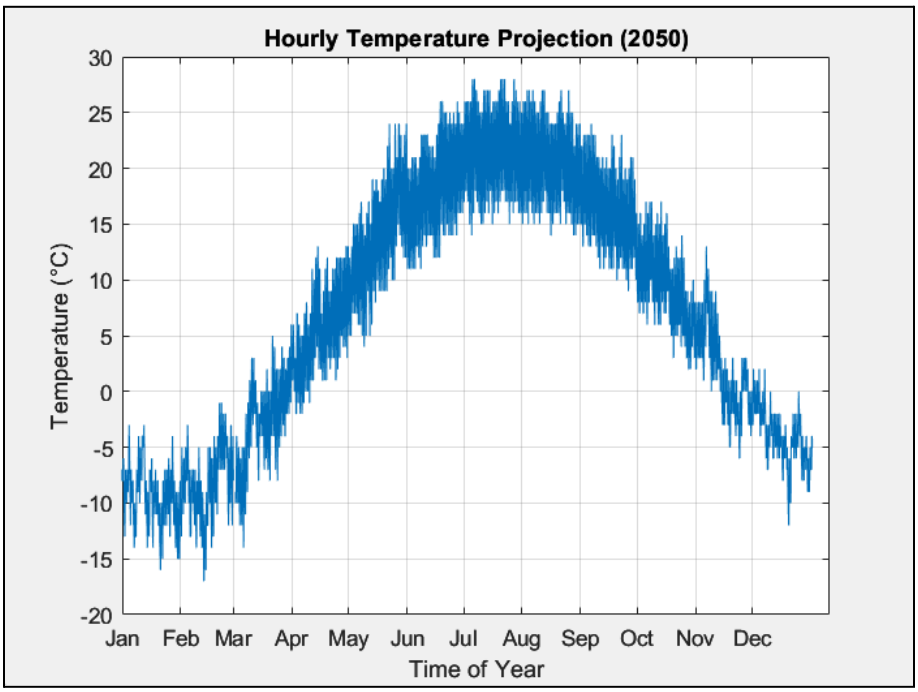


Figure 4: Hourly temperature projection (2050)

2.4: Ottawa commute distances

I contacted City officials to get accurate driving distance data for the city of Ottawa. I used a 2011 origin-destination survey to find the percentile breakdown of daily driving distances in Ottawa. [10]. The report breaks up Ottawa-Gatineau into the following neighbourhoods, with each neighbourhood considered an origin and destination. A map from the origin-destination study [10] is shown below in Figure 5; only the neighbourhoods in Ottawa are being considered.

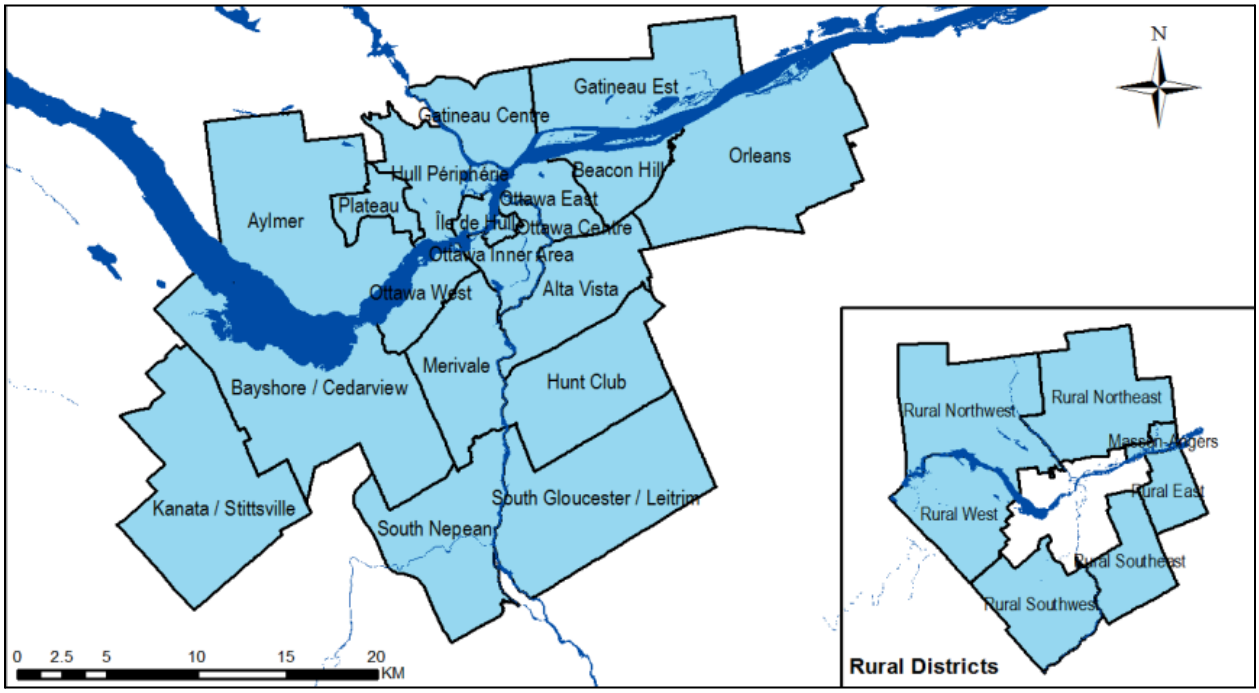


Figure 5: Origin-destination Ottawa breakdown

The 2011 survey contains the number of daily trips for all purposes from each origin to each destination. This data is in the appendix of this report.

I used QGIS to overlay the above maps over the city of Ottawa and find the centroids of each sub-area and the corresponding coordinates. Subsequently, the distance between each origin and destination was found. I used a Google Distance Matrix API key and a Python script that I wrote to find the distance by road between each sub-area. This script and all corresponding data are in the appendix of this report.

To find the distance of travel for trips within a sub-area, I used the square root of the area of each sub-area.

With a MATLAB script, I found the number of trips and their distance using the number of trips from each origin to each destination and the corresponding distances. This same MATLAB script was used to find a percentile breakdown for the distance of trips in Ottawa; this is shown in Figure 6.

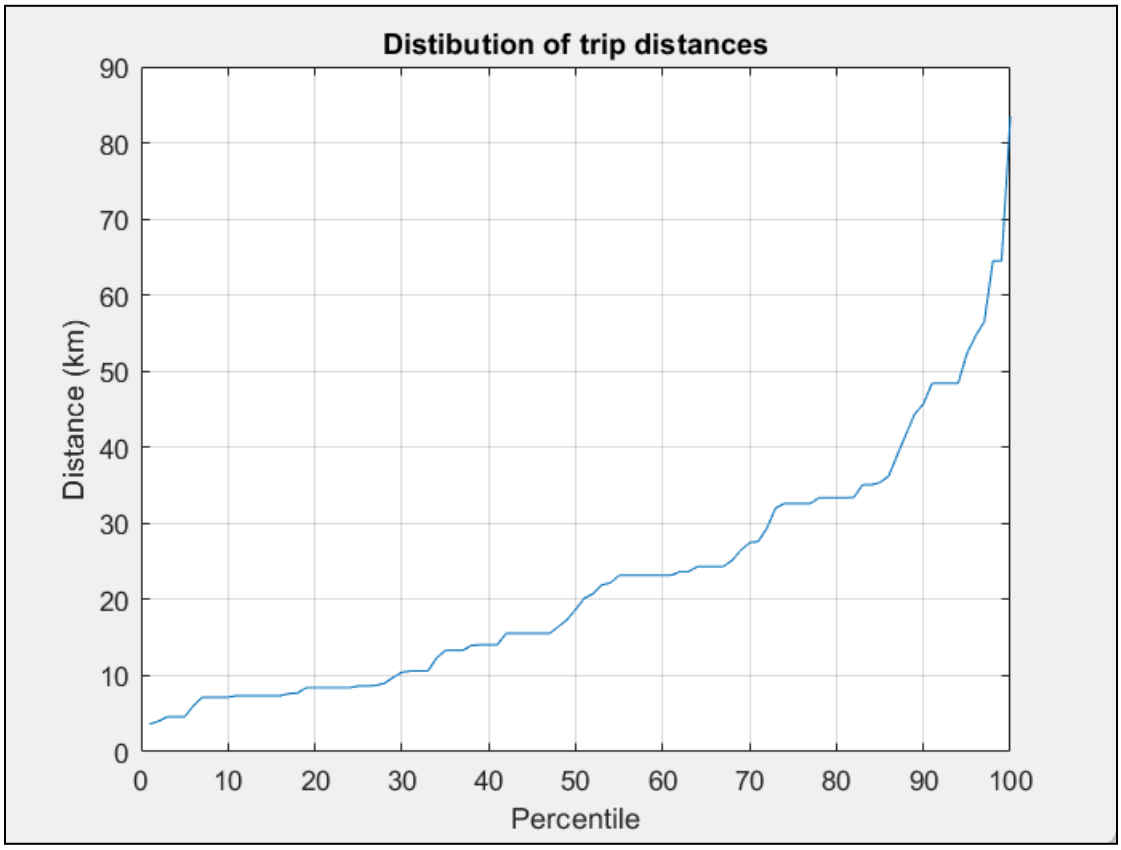


Figure 6: Ottawa driving distance percentile breakdown.

I used this breakdown when finding the projected hourly load. Furthermore, I used the Origin-Destination Survey [10] estimated number of trips per person to be 2.67; the distance values outlined above were therefore multiplied by the number of trips travelled. This method assumes that even those with the greatest driving distances still make 2.67 trips a day; in this regard, refinement could be made to include the difference in the number of trips per person, as it relates to their commute distances. As it is expected drivers who have the largest commutes are making fewer trips than those with smaller commute distances.

This data was also collected over the fall; it assumes that the number of trips is an average of all the data accumulated, making it a good representation of both weekday and weekend driving habits. It was also assumed that the trips made to not vary with seasons. Trips starting or ending outside of Ottawa were not used to find the percentile breakdown. The data currently used is also from the 2011 survey; changes in population in different subareas have occurred since then. The city is currently in the process of releasing the findings of the 2022 origin-destination survey. Results will be updated with new data as it becomes available.

2.5: MATLAB Script

All these inputs are imported into MATLAB as CSV's. When the script is run, it prompts the user to enter the number of EVs in the community, the percentage of workplace charging, and the maximum standard deviation in hours. (Workplace, Residential, and Weekend).

When run, the script will generate an hourly demand profile for one year and output it as a CSV. The figure below shows the projected EV demand for 2050, assuming 412,000 EVs, 15% workplace charging, and a max standard deviation of 2 hours. The script used is in the Appendix of this report.

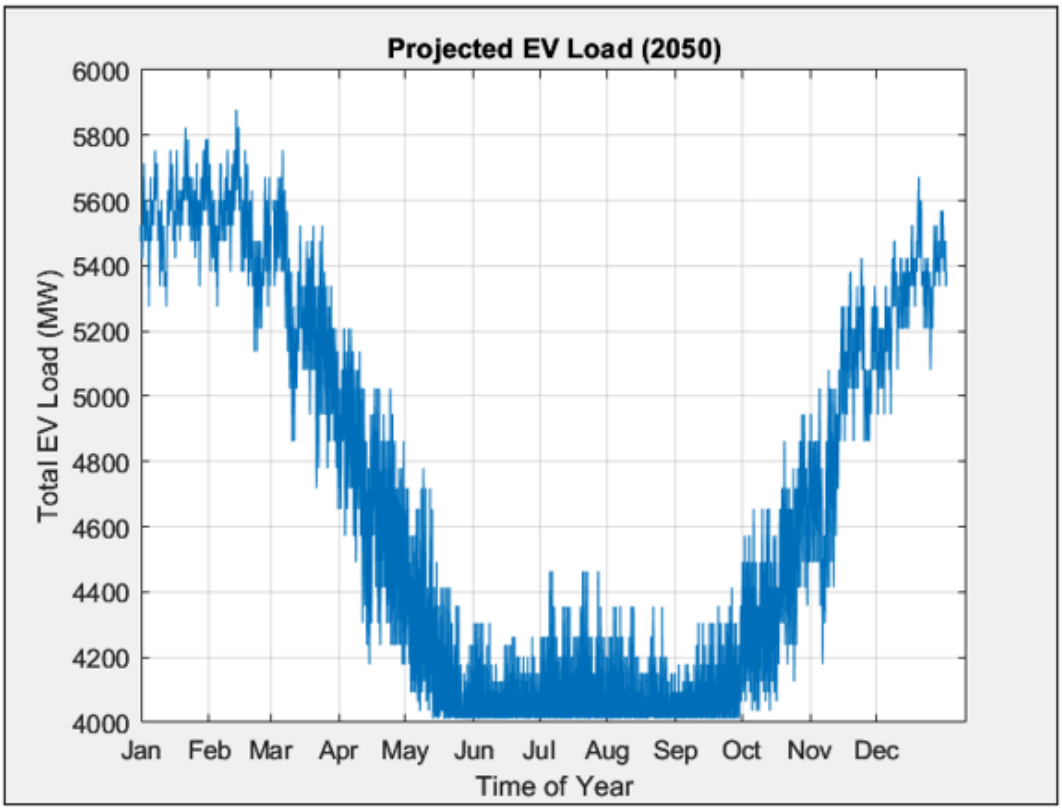


Figure 7: Projected EV Load (2050)

I estimate the peak demand to be 5,800 MW on February 13th.

3.0 Optimization Model

The primary objective of the integration team is to create a detailed optimization model that effectively coordinates the integration of various technologies investigated by each sub-team. The model aims to minimize costs, including the costs of generation, capital investment, and the lifespan of these technologies. The inputs for the model are twofold: data supplied by each sub-team and data independently gathered by the integration team.

Sub-teams have received guidance on the necessary data and its structure. As the year progresses, they will focus on their technologies' design and data collection, enabling successful incorporation into the overall integration model.

This term, the integration team has concentrated on accumulating data to project the 2050 demand in Ottawa, encompassing sectors like electric vehicles (EVs), residential, industrial, and commercial areas. Additionally, we factored in the costs related to energy imports, both financial and carbon.

3.1 MATLAB model

While the integration team plans to have the final optimization model built later, I built a "Dummy Model" in MATLAB as a starting point. The MATLAB code, in its entirety, is in the appendix of this report.

The parameters of the model are as follows.

```
% Parameters
minStateOfCharge = 20; % Minimum state of charge for storage
chargeRate = 10; % Charge rate of storage
dischargeRate = 10; % Discharge rate of storage
initialSoC = 25; % Initial state of charge
costStoragePerMW = 50; % Capital cost of storage per MW
costStoragePerMWh = 80; % Operational cost of storage per MWh

costImports = 90; % Cost of imports per MWh

costWindPerMW = 1000; % Capital cost of wind per MW
costSolarPerMW = 1200; % Capital cost of solar per MW
costNuclearPerMW = 1500; % Capital cost of nuclear per MW

costWindPerMWh = 20; % Operational cost of wind per MWh
costSolarPerMWh = 60; % Operational cost of solar per MWh
costNuclearPerMWh = 70; % Operational cost of nuclear per MWh

maxStorageCapacity = 200; % Maximum storage capacity
maxWindCapacity = 100; % Maximum Wind capacity
maxSolarCapacity = 100; % Maximum Solar capacity
maxNuclearCapacity = 100; % Maximum Nuclear capacity
```

Figure 8: MATLAB dummy model placeholder parameters

At this point in the project, I use placeholder values for various parameters for the generation and storage sub-teams. These inputs will be refined and updated as the sub-teams provide us with their data. The integration team will find the associated cost of energy imports.

The model attempts to accurately calculate the hourly energy output from each generation technology, along with monitoring the energy storage system, including its state of charge and whether it is in a charging or discharging state. We accomplish this by optimizing for the most cost-effective solution, considering both the generation cost (in $/MWh) and the capital cost associated with the peak generation capacity of each technology (in $/MW). The model ensures that the energy demand is consistently met every hour while adhering to predefined constraints.

The model effectively determines the optimal solution, factoring in total cost, hourly generation profiles, and the hourly state of charge, including the charging/discharging rate of the energy storage systems. This model serves as a foundational tool for further development, with planned future enhancements detailed in the preceding section.

4.0 Future Work

Charging Profile Validation: Future research may introduce alternative profiles. We may consider a public charging profile.

Time of Use Rates and Peak Usage: The residential and workplace profiles are based on current time-of-use rates, which show a residential charging peak at 9 PM. However, with increasing EV adoption, peak times may shift. We will conduct research into how these profiles might change. We may need to alter the residential charging profile so that its peak occurs during the night as EV adoption increases. Furthermore, we could research how smart charging among fleets of vehicles could affect these profiles.

Workplace Charging Assumption: The assumption that 15% of users charge at work during weekdays is variable in our MATLAB simulations, allowing for sensitivity analysis. We will refine this percentage as more data becomes available.

Model Work: The current model is a starting point that we will finalize by the end of the winter semester. By then, we will include the variability of the availability of wind and solar power. We will also factor in financing and the lag time it takes to get generation technologies online, as well as other considerations.

The methodology used to project the EV demand allows for projecting the demand yearly, from now until 2050; similarly, data used for other Load Profiles also allows this. Having this data will allow the model to run simulations over multiple years, considering financing, lifespans of technologies, and associated decommissioning/maintenance costs. This will create a more comprehensive model to simulate more realistic outcomes.

In the figure below a rough Gantt chart outlines the work for the winter semester. This is subject to change, however the bulk of the work in the winter semester should stay consistent

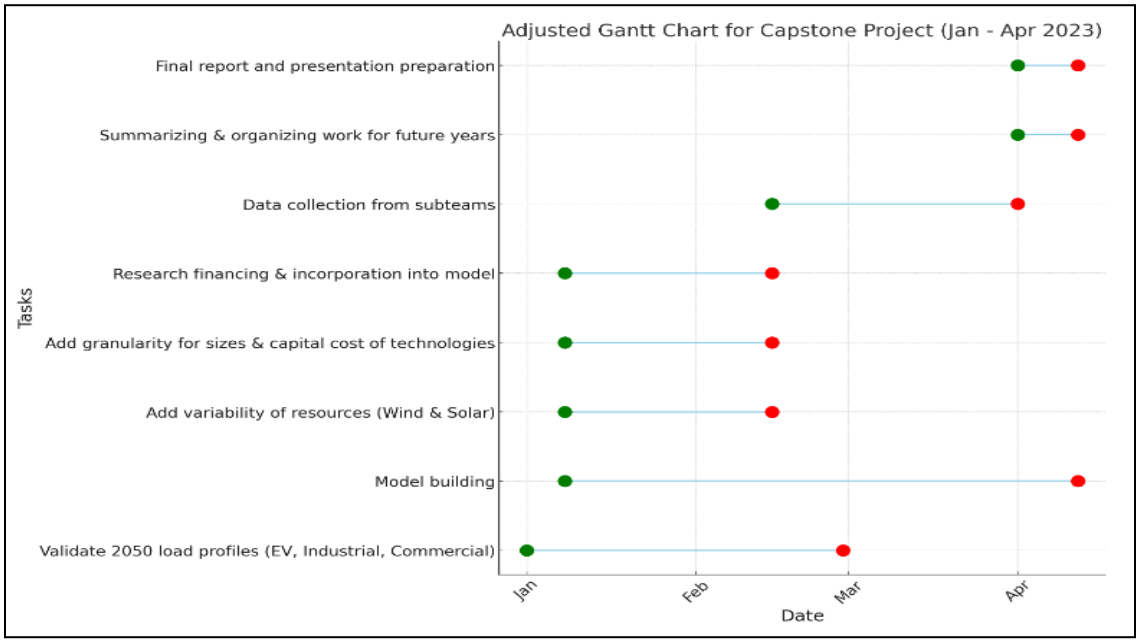


Figure 9: Gant chart for future work

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[11] D. Kriger, C. Chartrand, A. Rose, D. Cleghorn, K. Shen, and T. Erwin, "2011 TRANS O-D Survey Report," R.A. Malatest & Associates Ltd., Jan. 2013 Accessed: November 19, 2023].